CONFIDENTIAL

MASTER FORMULA AND METHOD

OF

METRONIDAZOLE INTRAVENOUS INFUSION BP 100ml

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BIOMEDICAL LIMITED

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Approval Page

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A. Product Information

METRONIDAZOLE INTRAVENOUS INFUSION BP 100ml

Brand Name : Bioflex Metronidazole Intravenous Injection 500mg/100 ml

Generic Name : Metronidazole Intravenous Infusion BP

Product Category : Antibacterial

Dosage Form : Intravenous Infusion

Route of Administration : Intravenous

Label Claim : Each 100 ml Contains…

Metronidazole	BP	500mg
Sodium Chloride	BP	0.74gm
Water for Injection	BP	q.s. to 100 ml

Packing Pattern : 40 pieces of over wrapped pouches packed in a mono carton and placed in 5-ply cartons along with the handling and storage label. The cartons are sealed with gum or starch paste.

Shelf Life : 36 Months.

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C. Manufacturing Formula

Standard Batch Size = 1100 Liters.
Standard Fill Volume = 100ml per PVC Pouches.

Sr. No.	Raw Materials	Rational	Label claim/100ml	% Ove-rages	Std. Batch Qty.
1	Metronidazole Powder BP	Active	500mg	Nil	5.50 Kg
2	Sodium Chloride BP	Excipient	740mg	Nil	8.14 Kg
3	Water for Injection BP	Vehicle	N.A.	Nil	q.s. to 1100 Liters.

PRIMARY PACKING MATERIAL

Sr. No.	Name of the Material	Unit	Batch Qty.	% Excess	Std. Batch Qty
1	100ml PVC Pouches	Nos.	10,050	0.5%	10,000
2.	Stoppers	Nos.	10,100	1 %	10,000

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B. Manufacturing Conditions and Precautions

1. Strictly follow the procedures of Entry/Exit to Clean Areas.
2. Ensure that the manufacturing area and equipment are clean.
3. Ensure that at all the time current good manufacturing practices are followed.
4. Follow safety measures all the time.
5. Ensure that the weighing scale is calibrated.
6. Verify the weight of dispensed raw material.
7. Rinse the equipment with fresh water for injection before every use.
8. Check the manufacturing area & Filling area for pre-operation checks like valve opening and closing.
9. Destroy the rejected PVC pouches as per the procedure.
10. Avoid product exposure to dirt during manufacturing, filling and storage.

SR. No	NAME OF EQUIPMENT	IDENTIFICATION No.	QUANTITY	CAPACITY
1	WEIGHING BALANCE	BL/WB/PE/001	1	50kg
2	DISPENSING BOOTH	BL/DSP/PE/001	1	FITTED WITH 0.2µ HEPA FILTER
3	MIXING TANK	BL /SPR/PE/OO7-3	1	2300 LITERS
4	MIXING TANK	BL /SPR/PE/OO7-2	1	3000 LITERS
5	MIXING TANK	BL /SPR/PE/OO7-1	1	3300 LITERS
6	SEMI AUTOMATIC FILLING MACHINE	BL/FR/PE/001 – 3	3	0 -2000ML
7	SEMI AUTOMATIC FILLING MACHINE	BL/FR/PE/004 – 5	2	0 -2000ML
8	DE-LAMA AUTOCLAVE (STERILISER)	BL/AC/PM/001	4	3,200Litres
9	CLEAN ROOM	BL/CL/PE/001	1	FITTED WITH 0.2µ HEPA FILTER
10	POUCH CODING MACHINE	BL/BC/PE/001-004	4	NA

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E. Manufacturing Procedures

1. Samples of Water for Injection. B.P. are collected and send for analysis to the Quality Assurance department. If the Quality assurance certifies the water complies for manufacturing, proceed to step 2.
2. Turn on the ball valve the open WFI into the tank, switch on WFI transfer pump and allow WFI to run from the storage vessel into the calibrated manufacturing vessel up to 1000 liters of the required volume.
3. Weigh 8.14kg of Sodium Chloride B.P. and 5.5Kg of Metronidazole Powder BP on the pre calibrated Weighing scale under the dispensing booth in the presence of Production Pharmacist and QA officer.
4. Transfer the dispensed Raw materials into the manufacturing Tank.
5. Note the time and record it.
6. Turn on the stirrer and mix thoroughly, gradually making up the volume to 1200Liters.
7. Continue stirring for 10 – 15minutes to dissolve the ingredient.
8. Inform the Quality Assurance so that samples can be collected for analysis.
9. If Quality Assurance has certified the bulk solution to be within the compendia specification, intimate the filling section so that filling can commence.
10. Ensure solution transfer pump is switched “ON”
11. Fix the sterile filter membranes in the filter holding unit
12. Fix the 0.8µ pre filter in filter holder and 0.2µ cartridge filter into its holder and ensure that there are no leakages of solution during filtration.
13. Check the correctness of the printing details of the pouches. Record the filtration and filling details in the Batch Record.
14. Adjust the fill volume 105ml for 100ml pouch filling.
15. Rinse the stoppers in WFI and keep near the filling machine
16. Switch on the filling machine and fill the required volume in PVC pouches and insert Stopper inside the open neck of the PVC pouch and pass through the transfer box to loading area for sterilization
17. Check the fill volume by measuring with a measuring cylinder the volume of the filled pouch every 10 minutes from the filling nozzles and record it in Batch Record.
18. Follow proper cleaning procedure batch to batch and product to product cleaning.
19. Sterilize the filled pouches in super-heated water spray sterilizer at 116°C-118°C for 40 min
20. QA will take 6 samples for finished product analysis from the sterilized product on the trolleys.
21. Transfer the batch for visual inspection and packing.

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F. In process Control

1. Once mixing is complete the production supervisor will notify the QC for sampling
2. Sample 250ml of the solution
3. Check for clarity, PH, and assay of solution as per standard testing procedure for dextrose saline solution
4. Record the in-process analysis results in the BMR and the record in the in-process analysis record sheet for metronidazole.
5. Notify the production supervisor of the result of the analysis as soon as it passes for them to commence filtration and filling process.
6. Check the correctness of the pouches before filling. Check for the product name, batch coding details, manufacturing and expiry date
7. Check the fill volume from each of the nozzles at the start of filling and every 10 minutes afterwards

G. Packaging Formula

Standard Batch Size = 1100 Liters for 100ml pack size.

Sr. No.	Name of material	Unit	Quantity Required	% Ove-rages	Std. Batch Qty.
1	Nylon wrappers 500 ml	No	10,100	1 %	10,000*
2	Infusion Carton	No	250	Nil	250
3	Stickers	No	250	Nil	250

***** Subjected to variation depending on yield at visual inspection stage.

Packing Pattern: 40 pieces of over wrapped pouches packed a mono carton and placed in 5-ply cartons along with the handling and storage label. The cartons are sealed with gum or starch paste.

H. Packaging Conditions and Precautions.

1. Check the cleanliness of area & equipment.
2. Check for line clearance & record the details in Batch Record.
3. Ensure that all packing materials required for a batch are requested.
4. Verify all printed matters on the pouch such as:
   i) Batch No.
   ii) Mfg. Date.
   iii) Exp. Date.
5. Ensure withdrawal of retention sample.

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I. Packaging Procedure

(Visual Inspection and packing is a simultaneous continuous operation)

1. Perform the pre-operation checks. Ensure for Q.A. release is obtained.
2. Check for the line for proper arrangement of workers.
3. On the carton check the proper marking roller fix the required digits and alphabets stating product name, batch number, Mfg. Date & Exp. date.
4. Take one trolley of the batch to the packing line.
5. Intimate the In Process Quality Assurance Inspector for the line clearance.
6. In Process Quality Assurance Inspector will check for the line clearance & sign in the specified place in the BMR.
7. Start coding of cartons & packing of batch.
8. Receive the batch from Visual inspection area (On line visual inspection and packing)
9. Put the pouches to LDPE bags with the tube side of the pouch to the inner side of the bag without tearing the bag.
10. Transfer the wrapped pouches to the cartooning area.
11. Put the wrapped pouches in the cartons on which a product identification sticker label has been affixed.
12. Arrange 40 pouches each in a mono carton in a 5-ply carton.
13. The arrangement of pouches in the batch coded carton having product identification sticker affixed should be of uniform pattern & should be placed flatly in the cartons.
14. Seal the cartons with gum or starch paste freshly prepared. Apply the gum or starch paste with a brush.
15. 20 pouches for QA retention samples per batch taken for reference.

J. Disposal of Rejects.

1. Any rejected pouches from the batch coding, Filling, Sterilization and Packing sections will be collected by the supervisor every shift & will get approval from Quality Assurance Manager and Superintendent pharmacist for destruction on batch wise.
2. The Supervisor when handing over the rejected pouches for destruction will write the details of the rejections in the record & get it countersigned by the reject destruction personnel as well as QA personnel.
3. The rejected pouches are slitted at least to a span of 3 inches with the help of blade scissors & drained out if there is any solution remaining in the pouch. This operation will be carried out on direct supervision of the QA personnel.
4. After this slitted pouches are packed into LDPE bags and transferred to scrap yard for sale to the Authorized scrap collection contractor.

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K. Yield Analysis

Standard Batch size = 1100 Liters.

Sr. No.	Stage	Standard Yield %	Permissible Yield %
1.	Batch Coding	100%	NLT 99.50%
2.	Filling	100%	NLT 99.50 %
3.	Packing	100%	NLT 94%

L Reprocessing

Batch reprocessing is not practiced.

M Process Deviation

Any inevitable deviation should not be carried out without approval of Quality Assurance Manager and authorization of Superintendent Pharmacist. Record any process deviation in a specified format in the deviation form and attach same to the BMR.

N Raw Material Specification

1. Metronidazole Powder (As per British Pharmacopoeia)
2. Sodium Chloride (As per British Pharmacopoeia)

O Packing Material Specification

1. PVC pouches
2. Stoppers
3. 100ml Nylon Wrappers
4. 100ml Cartons
5. Gum or Starch paste
6. Infusion leaflets/Labels

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Q Finish Product Specification

QC Specification (As per British Pharmacopoeia)

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